

Orthogonal error spaces for two ordered product quadratures

Auxiliary exact identity; independently reviewed

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The sharp simultaneous two-die question remains open in this work. This note reformulates its real hybrid version as orthogonality of cumulative quadrature errors. It is not a new upper or lower size bound.

Let μ_1, μ_2 be probability measures on $[0, 1]$ with no common atoms, and put $\delta_i = \mu_i - dt$. For a polynomial f define

$$E_{i,f}(t) = \int_{[0,t]} f(x) d\delta_i(x).$$

Endpoint conventions do not affect the Lebesgue integrals below.

Theorem 1. *Suppose each μ_i integrates polynomials through degree $M + 1$ exactly as Lebesgue measure, and the product rule integrates every polynomial of total degree at most M exactly on each open chamber $x < y$ and $y < x$. Then*

$$\int_0^1 E_{1,f}(t) E_{2,g}(t) p(t) dt = 0 \quad \text{if } \deg f + \deg g + \deg p \leq M - 1. \quad (1)$$

In particular the two spaces

$$\mathcal{E}_i^{(r)} = \{E_{i,f} : \deg f \leq r\} \subset L^2[0, 1]$$

are orthogonal whenever $2r \leq M - 1$. If μ_i is finitely supported, each of these spaces has dimension exactly $r + 1$.

Proof. For $\deg f + \deg g \leq M$, expand the ordered product error as

$$C(f, g) = \int_{x < y} f(x) g(y) d\delta_1(x) d\delta_2(y).$$

Its $\mu_1 \otimes \mu_2$ and Lebesgue product terms are equal by the chamber assumption. Each mixed term also equals the Lebesgue integral: for example integrating $g(y)$ from x to 1 leaves a polynomial in x of degree at most $\deg f + \deg g + 1 \leq M + 1$, to which the first marginal rule applies. Hence $C(f, g) = 0$. The same argument applies to the reverse chamber and to every pair of polynomials within the degree budget. Since the measures have no common atoms, their cumulative error functions have no simultaneous jumps. It follows that

$$\int E_{1,f} dE_{2,g} = 0, \quad \int E_{2,g} dE_{1,f} = 0 \quad (\deg f + \deg g \leq M).$$

Let h be an antiderivative of p . Stieltjes integration by parts for $hE_{1,f}E_{2,g}$ has no common-jump term. Its boundary term is zero: cumulative errors at 1 vanish by marginal exactness, and at 0 any endpoint masses cannot occur simultaneously. Therefore

$$\int p E_{1,f} E_{2,g} = - \int E_{1,f} dE_{2,hg} - \int E_{2,g} dE_{1,hf} = 0,$$

because $\deg f + \deg g + \deg h \leq M$. This proves (1). For a finitely supported μ_i , on the complement of its finitely many atoms the derivative of $E_{i,f}$ is $-f$. Thus $E_{i,f} = 0$ almost everywhere implies $f = 0$, proving the dimension statement. $\square$

Equivalent starting condition. Under marginal exactness, the chamber condition is equivalent to $\int E_{1,f} dE_{2,g} = 0$ for all $\deg f + \deg g \leq M$; expanding $\delta_1 \otimes \delta_2$ proves both implications. The Lebesgue orthogonality in (1) is a consequence, not asserted to be sufficient by itself.

Relevance and limitation. Two small real discrete variables together with u independent uniforms have scalar marginal exactness through degree u and chamber exactness through degree u (the conditional-order kernels form a Bernstein basis on each triangle). Thus the theorem applies with $M = u - 1$, giving (1) through total degree $u - 2$. The extra marginal degree in the theorem should not be inferred from chamber degree alone. Consequently matching their scalar quadrature moments separately is insufficient. The two cumulative-error spaces must also be orthogonal. The reviewed polynomial construction achieves this by positive density completion. No argument here reduces its counts to $O(M^2)$, nor rules out that sharper possibility.